Methods | MTH1002 | Term 2
Dr. Mike O'Sullivan | Year: 2025-2026

Exercise Sheet 1 (Tutorials in week 2)

Key:

- **Questions marked with a ☕** are for discussion in your tutorials. **Please complete these questions before your tutorial.**
- **Questions marked with a 🦇** are difficult questions.
- **Questions marked with a ♫** have been marked so that they don't feel left out. Please don't leave them out during your independent learning and revision time.

☕ **Exercise 1.** In general, matrix multiplication is not commutative. Suppose that the matrix $A = (a_{ij})_{2\times 2}$ is

$$A = \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix}.$$

Find the general form of the matrix $X = (x_{ij})_{2\times 2}$ such that $AX = XA$.

☕ **Exercise 2.** Compute the determinant of the matrix $A = (a_{ij})_{4\times 4}$ given by

$$A = \begin{bmatrix} 2 & 4 & -6 & 8 \\ 1 & -2 & 3 & -4 \\ 3 & 1 & -2 & 5 \\ 1 & -1 & 2 & 3 \end{bmatrix}.$$

♫ **Exercise 3.** Determine the rank of $A = (a_{ij})_{4\times 5}$ given by

$$A = \begin{bmatrix} 3 & 2 & 0 & 5 & 0 \\ 3 & -2 & 3 & 6 & -1 \\ 2 & 0 & 1 & 5 & -3 \\ 1 & 6 & -4 & -1 & 4 \end{bmatrix}.$$

♫ **Exercise 4.** Consider a system of linear equations in the form

$$A = \begin{bmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 3b & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 4 \end{bmatrix}.$$

Determine values of a and b such that the system has (i) no solutions, (ii) a unique solution, and (iii) an infinite number of solutions. Find the solutions for cases (ii) and (iii).

☕ **Exercise 5.** Find all the eigenvalues and eigenvectors of the matrix $A = (a_{ij})_{3\times 3}$ in the form

$$A = \begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix}$$

by solving first $\det(A - \lambda I) = 0$ and, then, $Av = \lambda v$.

♫ **Exercise 6.** Given the four vectors $v_1 = (1, 3, 0, 5)^T$, $v_2 = (1, 2, 1, 4)^T$, $v_3 = (1, 1, 2, 3)^T$ and $v_4 = (1, -3, 6, -1)^T$, find the value of λ in the vector $w = (1, 0, \lambda, 2)^T$ such that w can be expressed as a combination of the four vectors (i.e., the vectors w, v_1, v_2, v_3, v_4 are linearly dependent).

♫ **Exercise 7.** Compute the determinant of the matrix $A = (a_{ij})_{5\times 5}$ given by

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & (\alpha+1) & 3 & 4 & 5 \\ 1 & 2 & (\alpha+1) & 4 & 5 \\ 1 & 2 & 3 & (\alpha+1) & 5 \\ 1 & 2 & 3 & 4 & (\alpha+1) \end{bmatrix},$$

where α is a nonzero number.

♫ **Exercise 8.** Suppose that $A = (a_{ij})_{n\times n}, B = (b_{ij})_{n\times n}$ and $C = (c_{ij})_{n\times n}$ and satisfy

$$AB = BC = CA = I$$

where I is the $n \times n$ identity matrix. Find $A^2 + B^2 + C^2$.

♫ **Exercise 9.** If the three vectors, v_1, v_2 and v_3, are linearly independent, show that $(v_1 + v_2)$, $(3v_2 + 2v_3)$ and $(v_1 - 2v_2 + v_3)$ are also linearly independent.

♫ **Exercise 10.** Suppose that $\lambda_1 = 0$ is an eigenvalue of the matrix $A = (a_{ij})_{3\times 3}$ given by

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & \alpha \end{bmatrix}.$$

Determine the value of α and find the other two eigenvalues λ_2 and λ_3.

♫ **Exercise 11.** Consider the equation

$$Ax = y,$$

where

$$A = \begin{bmatrix} 1 & -1 & -1 \\ 2 & -1 & -3 \\ -3 & 4 & 4 \end{bmatrix}, \; x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \; y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}.$$

(i) Determine the value of $\det A$; (ii) derive the inverse matrix A^{-1}; and (iii) express (x_1, x_2, x_3) in terms of (y_1, y_2, y_3) (such as $x_1 = 4y_1 + y_3$).